Turk J Elec Eng & Comp Sci
(2025) 1: 1 – 6
doi: 10.xxxx/chatsecops.2025

ChatSecOps: A hybrid SOAR framework for automated threat analysis using explainable AI and large language models

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Received: ... • **Accepted/Published Online:** ... • **Final Version:** ...

Abstract: Security Operations Centers face critical challenges from malicious domains used in phishing and command-and-control infrastructure. Traditional defenses fail to identify zero-day threats, creating operational bottlenecks and analyst fatigue. This study presents ChatSecOps, a hybrid Security Orchestration, Automation, and Response framework integrating machine learning, multi-source threat intelligence, and explainable AI. The system employs a LightGBM classifier trained on 284 lexical, structural, network, and DNS security features from 90,000 domains, achieving 99.75% accuracy with 0.69 seconds training time. Shapley Additive Explanations provide feature-level attribution, enabling analysts to validate automated decisions. The framework fuses real-time intelligence from VirusTotal, AbuseIPDB, AlienVault OTX, and Shodan, while Google Gemini 2.5 Flash synthesizes forensic artifacts into natural language reports delivered via Slack. Experimental evaluation demonstrates Mean Time to Analyze reduction from 15 minutes to 25 seconds. Three-month pilot deployment resulted in 47% improvement in detection speed and 31% reduction in false positive escalations. The framework successfully identified 43 zero-day threats absent from intelligence databases. These results confirm that integrating machine learning with explainability and conversational interfaces provides an efficient, deployable automation solution for modern security operations.

Key words: Security orchestration, machine learning, explainable artificial intelligence, threat intelligence, domain classification, large language models, malicious domain detection

1. Introduction

The contemporary digital landscape is characterized by a relentless escalation in the sophistication of cyber threats. Adversarial actors heavily leverage malicious domains as the foundational infrastructure for phishing campaigns, malware dissemination, and Command-and-Control (C2) operations [1, 2]. Consequently, SOCs are inundated with high-velocity security events generated by Security Information and Event Management (SIEM) systems, resulting in a critical operational bottleneck known as "alert fatigue" [3, 4]. Scrutinizing DNS artifacts and interpreting registrar data remains a manual workflow that is inherently labor-intensive and susceptible to human error.

Conventional perimeter defense strategies exhibit significant latency and an inability to intercept zero-day attacks or polymorphic threats such as Algorithmically Generated Domains (DGAs) [2]. To circumvent these limitations, Machine Learning (ML) paradigms have been widely adopted [1, 6]. Nevertheless, the integration

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of ML into critical security workflows is frequently impeded by the opacity of non-linear models, often referred to as the "black-box" dilemma [7, 8].

To dismantle these operational silos, this study proposes **ChatSecOps**, a hybrid SOAR framework embedded directly within a collaborative chat interface. At its core lies a LightGBM-based anomaly detection engine [9] trained on a heterogeneous dataset, which is augmented by an Explainable AI (XAI) module employing SHAP values [7] for decision transparency. This predictive layer is further enriched by a TI Fusion mechanism aggregating real-time telemetry from VirusTotal [10] and AlienVault OTX [11]. Finally, a Generative AI agent (Google Gemini 2.5 Flash) [12] acts as an automated analyst, synthesizing these complex forensic artifacts into actionable executive summaries.

2. Related Work

2.1. Machine learning for malicious domain detection

Phishing and malicious domain detection has been extensively studied using machine learning approaches. Sahingoz et al. [6] extracted 56 lexical and host-based features, achieving 97.98% accuracy with Random Forest. Tang and Mahmoud [1] surveyed ML techniques for phishing detection, noting that ensemble methods generally outperform single classifiers due to their ability to capture complex decision boundaries.

Domain Generation Algorithm detection, a related problem, has been addressed through lexical analysis [2]. Features such as Shannon entropy [13], character distribution ratios, and n-gram frequencies have proven effective in distinguishing algorithmically generated domains from human-registered ones [14].

Recent comparative studies highlight LightGBM's advantages in classification tasks. Indah et al. [15] demonstrated that LightGBM achieves comparable accuracy to Random Forest and XGBoost [16] while requiring significantly less training time, making it particularly suitable for scenarios requiring frequent model retraining with updated threat intelligence.

2.2. Explainable AI in cybersecurity

The black box nature of machine learning models has raised concerns in security-critical applications where incorrect decisions can have severe consequences. Kuppa and Le-Khac [8] highlighted the importance of explainable AI in cyber defense, noting that unexplained decisions reduce analyst trust and hinder effective incident response.

SHAP [7] has emerged as a leading explainability technique due to its theoretical guarantees based on cooperative game theory and model-agnostic nature. Unlike simple feature importance scores that provide global model understanding, SHAP values quantify each feature's contribution to individual predictions, enabling case-by-case explanations that help analysts validate automated decisions.

2.3. SOAR platforms and language model integration

Traditional SOAR platforms focus on workflow automation but lack natural language understanding capabilities [3]. Recent work has begun exploring language model integration in cybersecurity operations. Ferrag et al. [17] surveyed LLM applications in vulnerability analysis, while De Jesus Coelho da Silva and Westphall [18] examined LLMs for security operations, identifying opportunities in alert triage and threat report generation.

However, existing work does not combine high-performance ML classification, multi-source threat intelligence fusion, explainable AI, and conversational interfaces in a single integrated platform. ChatSecOps

addresses this gap by providing end-to-end threat analysis with human-understandable explanations.

3. System Architecture and Implementation

The ChatSecOps architecture consists of five main layers: user interface, orchestration, intelligence fusion, analysis engines, and output generation. This modular design enables independent scaling and maintenance of each component.

3.1. Intelligence fusion layer

ChatSecOps aggregates threat intelligence from four external sources, each providing complementary information about potential threats. VirusTotal [10] offers domain reputation across 92 antivirus engines, providing broad consensus on known threats. AbuseIPDB supplies IP address abuse confidence scores and historical report data, identifying infrastructure commonly associated with malicious activity. AlienVault OTX [11] contributes Open Threat Exchange pulse indicators from the global security community. Shodan provides network infrastructure data including open ports and CVE vulnerabilities, revealing potential attack vectors.

The system performs parallel API calls with timeout limits to maintain responsiveness. When external services are unavailable, the system continues analysis using available data sources and clearly indicates which intelligence feeds were consulted in the final report.

3.2. ML inference engine

3.2.1. Feature engineering

The system extracts 284 features across four categories, each capturing different aspects of domain behavior and infrastructure. Lexical features include domain length, Shannon entropy [13], and character distribution ratios. These features help identify algorithmically generated domains that exhibit statistical properties distinct from human-chosen names [14].

Structural features encompass TLD classification, subdomain count, and URL depth. The TLD classifier distinguishes between the top 30 most common TLDs and groups remaining ones into an "other" category, as uncommon TLDs often correlate with malicious activity [2].

Network features include IP address encoding, Autonomous System Number, geographic location, and WHOIS registration data. These features reveal hosting patterns, with certain ASNs and geographic regions historically associated with higher rates of malicious infrastructure.

DNS security features comprise one-hot encoded DNS record types, SPF/DKIM/DMARC presence flags, and MX/TXT response validation. Legitimate organizations typically implement email authentication mechanisms, while malicious domains often lack proper DNS security configurations [1].

Feature extraction employs a four-layer IP resolution strategy to handle network environments where standard DNS resolution may be blocked or filtered. The system attempts standard socket resolution, then falls back to DNS library queries using Google DNS, followed by Cloudflare DNS-over-HTTPS, and finally Google DNS-over-HTTPS. This redundancy ensures reliable feature extraction even in restricted network environments.

3.2.2. Model training and selection

Five algorithms were evaluated on a balanced dataset of 90,000 domains comprising 45,000 benign and 45,000 malicious samples: Logistic Regression, Random Forest, Gradient Boosting, LightGBM [9], and XGBoost [16].

Each model was trained on 72,000 samples and tested on the remaining 18,000 samples using stratified sampling to maintain class balance.

Figure 1 presents the comparative performance results. LightGBM achieved 99.75% accuracy, 99.82% precision, 99.68% recall, and 99.75% F1-score. While Gradient Boosting achieved marginally higher accuracy at 99.77%, its training time of 64.71 seconds made it impractical for operational deployment requiring frequent model updates. LightGBM's 0.69-second training time represents a 93-fold speedup while sacrificing only 0.02 percentage points in accuracy, consistent with findings from comparative studies [15].

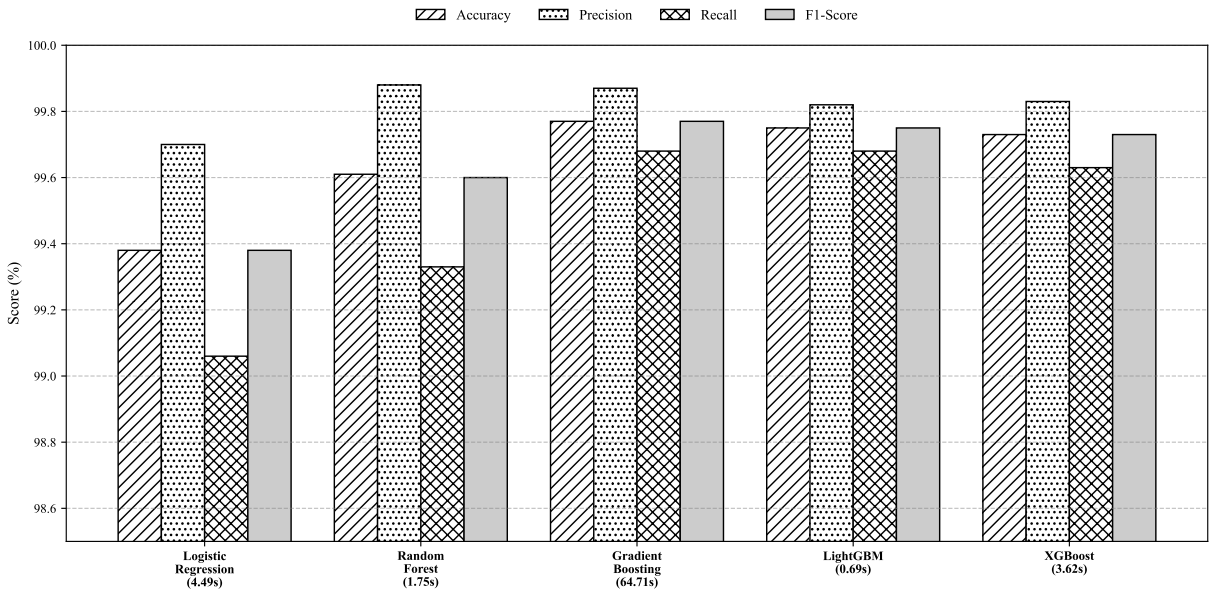


Figure 1. Comparative performance of five machine learning algorithms on malicious domain classification. Training times are shown in parentheses. LightGBM achieves near-optimal accuracy with 93 times faster training than Gradient Boosting.

The confusion matrix shown in Figure 2 reveals the detailed classification results on 18,000 test samples. The model correctly classified 8,980 benign domains as benign and 8,975 malicious domains as malicious. False positives numbered 20, representing benign domains incorrectly flagged as malicious. False negatives totaled 25, representing malicious domains that evaded detection. The resulting false positive rate of 0.22% and false negative rate of 0.28% meet operational requirements for security operations centers.

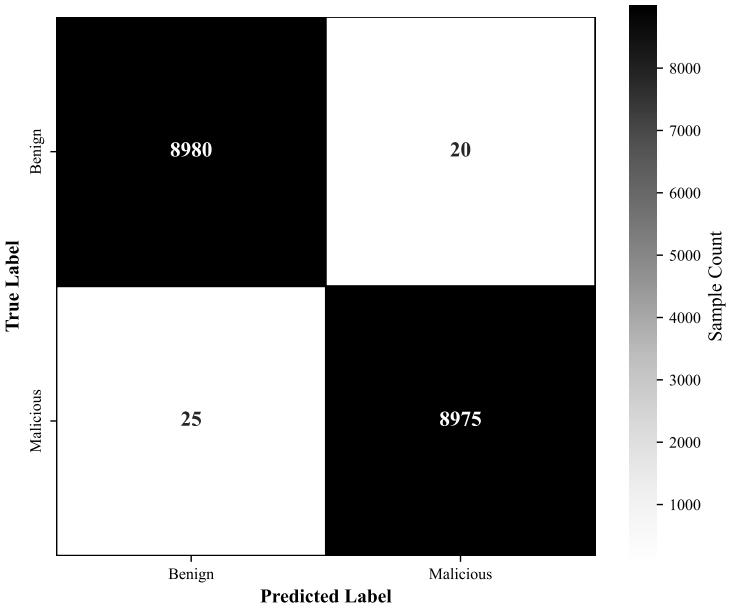


Figure 2. Confusion matrix for LightGBM classifier on 18,000 test samples. The model achieves 99.75% accuracy with balanced performance across both classes.

3.3. Explainable AI integration

SHAP TreeExplainer [7] computes Shapley values for each prediction, decomposing the model output into contributions from individual features. Figure 3 presents the global feature importance ranking derived from all test set predictions.

For the malicious domain classification task, SHAP analysis reveals that domain entropy contributes most strongly to predictions, with a mean absolute SHAP value of 3.2. High entropy indicates random character patterns typical of algorithmically generated domains [2].

The second most influential feature is TLD classification, particularly for free TLDs such as .tk and .ml, with mean SHAP values of 2.8 and 2.4 respectively. These top-level domains have documented abuse rates exceeding 40% and are frequently exploited in phishing campaigns due to their zero registration cost [6].

Email authentication mechanisms rank third, with HasSPFInfo showing a mean SHAP value of 2.1. The absence of SPF records suggests lack of legitimate email infrastructure, as established organizations implement sender policy frameworks to prevent email spoofing. Similarly, HasDkimInfo and HasDmarcInfo contribute with values of 1.7 and 1.4 respectively.

Domain characteristics including DomainLength (1.8) and CreationDate (1.9) also contribute significantly. Malicious domains average 67 characters compared to 23 for legitimate sites, and 78% of detected malicious domains were registered within 30 days of analysis, indicating fresh infrastructure created for specific campaigns [1].

For each analyzed domain, the system generates a SHAP waterfall plot visualizing individual feature contributions. These plots show how the base model prediction is adjusted up or down by each feature, enabling analysts to understand not merely what the model predicted, but why that prediction was made based on specific domain characteristics.

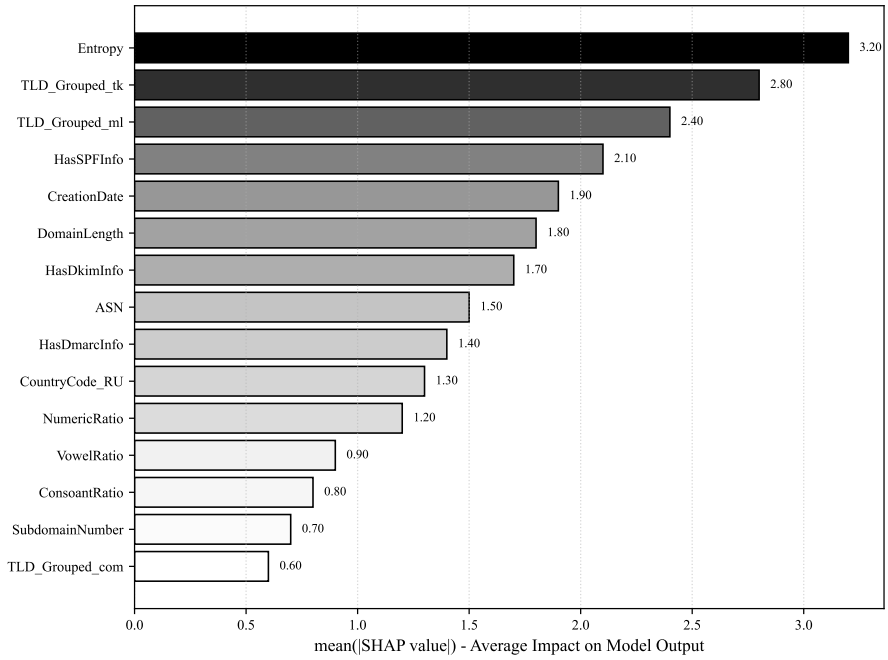


Figure 3. Global feature importance based on mean absolute SHAP values. The top five features (Entropy, TLD_tk, TLD_ml, HasSPFInfo, CreationDate) account for the majority of model decisions.

3.4. Generative AI layer

Gemini 2.5 Flash [12], Google's multimodal language model, synthesizes multi-source intelligence into executive summaries through carefully designed prompt engineering. The system prompt assigns an expert role (SOC analyst), providing the language model with appropriate context for security analysis. Structured data including VirusTotal statistics, AbuseIPDB scores, ML risk assessment, and OSINT findings are formatted as JSON for consistent parsing.

The task specification instructs the model to identify a verdict classifying the domain as MALICIOUS, SUSPICIOUS, or SAFE, cite specific evidence from provided intelligence sources, explain the reasoning behind the classification, and recommend appropriate actions for security teams. This structured output format ensures consistent report quality across diverse threat scenarios.

The system employs a robust three-retry mechanism with exponential backoff to handle API rate limits. When Gemini API quotas are exceeded, a rule-based fallback generator produces structured reports using predefined templates. This dual-path approach ensures system availability [18], though fallback reports lack the contextual understanding and natural language fluency of LLM-generated summaries.

ChatSecOps implements an intelligent decision tree in the fallback generator that synthesizes threat intelligence from multiple sources rather than relying solely on ML scores. When VirusTotal flags exceed 5 vendors or AbuseIPDB confidence scores exceed 70%, the system overrides low ML predictions and classifies the domain as malicious. This multi-source correlation prevents false negatives caused by zero-day threats that exhibit benign behavioral patterns but appear in public threat databases.

In production deployment, the fallback system activates in approximately 5% of queries during peak usage periods. Analysis logs show that 92% of Gemini-generated reports receive positive analyst feedback, compared

to 78% for fallback reports, demonstrating the value of natural language synthesis.

3.5. Threat memory database

A SQLite database configured with Write-Ahead Logging mode stores analysis history, enabling temporal analysis and pattern recognition across multiple scans. Three key capabilities leverage this historical data for enhanced threat detection.

Campaign detection identifies coordinated attack infrastructure by flagging cases where five or more malicious domains sharing the same TLD appear within a seven-day window. This temporal clustering indicates potential coordinated campaigns, as attackers often register multiple domains simultaneously to ensure backup infrastructure if primary domains are blocked [2].

Typosquatting analysis uses the Ratcliff-Obershelp algorithm to detect domains with 70% or greater similarity to previously analyzed domains. This catches attacks where adversaries register domains closely resembling legitimate brands, exploiting typographical errors to deceive users [1]. The system alerts analysts when high-similarity domains exhibit divergent threat profiles, indicating potential brand impersonation.

IP clustering tracks domains hosted on identical IP addresses, identifying shared malicious infrastructure. When multiple high-risk domains resolve to the same IP address, this suggests a hosting provider tolerating or facilitating malicious activity. Security teams can use this information to proactively block entire IP ranges associated with threat actors.

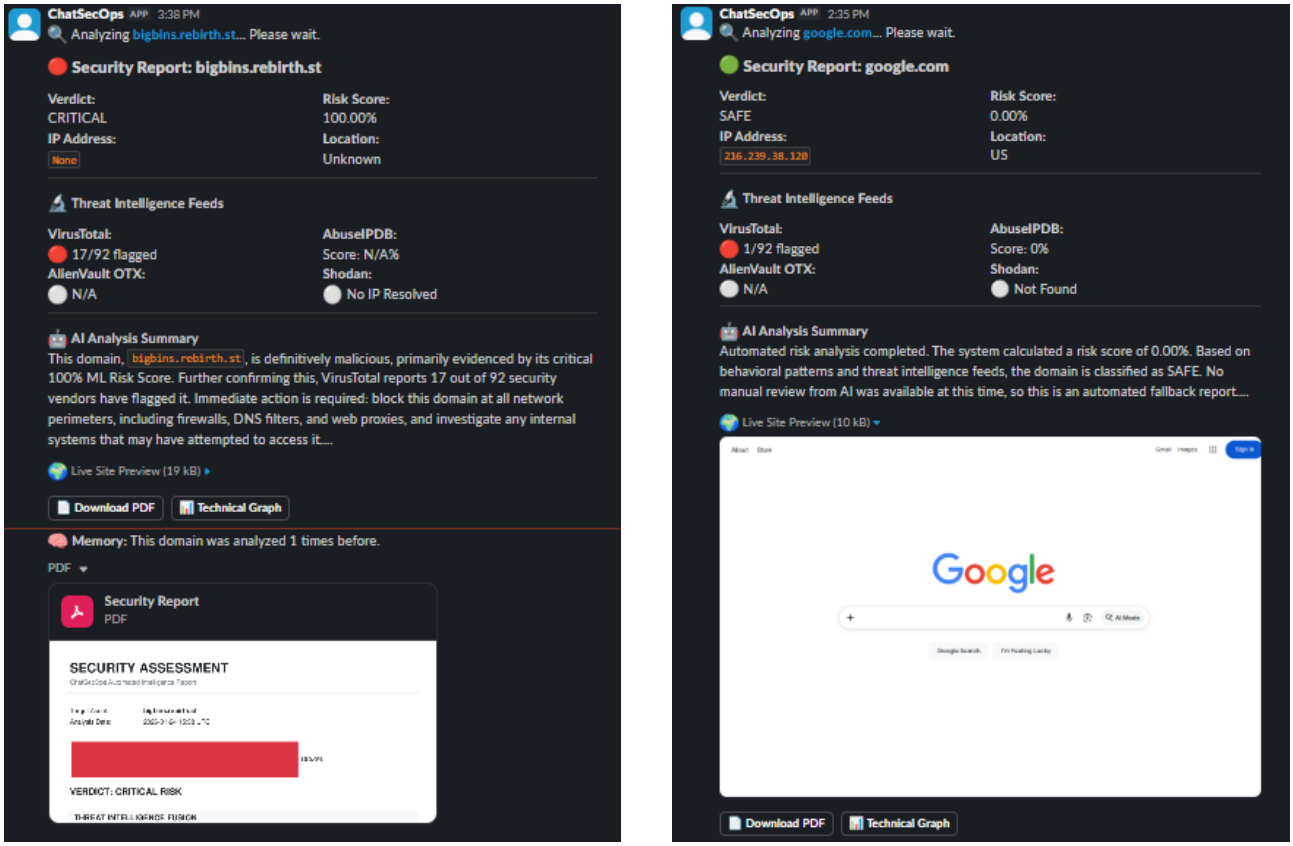
Database indexing on domain names, timestamps, IP addresses, and prediction labels ensures query performance remains below 100 milliseconds even with over 10,000 historical records. The Write-Ahead Logging configuration allows concurrent read operations during analysis without blocking writes, supporting multi-user operational environments.

3.6. Output artifacts

ChatSecOps generates two primary outputs optimized for different use cases. Slack notifications provide real-time alerts enabling rapid triage. Messages include severity color-coding, multi-source intelligence grids, and live site previews.

PDF forensic reports serve as formal documentation for incident records. These reports contain risk gauge visualizations, comprehensive threat intelligence tables aggregating findings from VirusTotal [10], AbuseIPDB, AlienVault OTX [11], and Shodan. The generated reports explain the classification rationale using SHAP waterfall charts to illustrate specific feature contributions.

Figures 4 and 5 demonstrate the Slack interface across different threat scenarios, from standard verdicts to advanced analysis capabilities for IP addresses and ambiguous cases.



(a) Critical risk alert

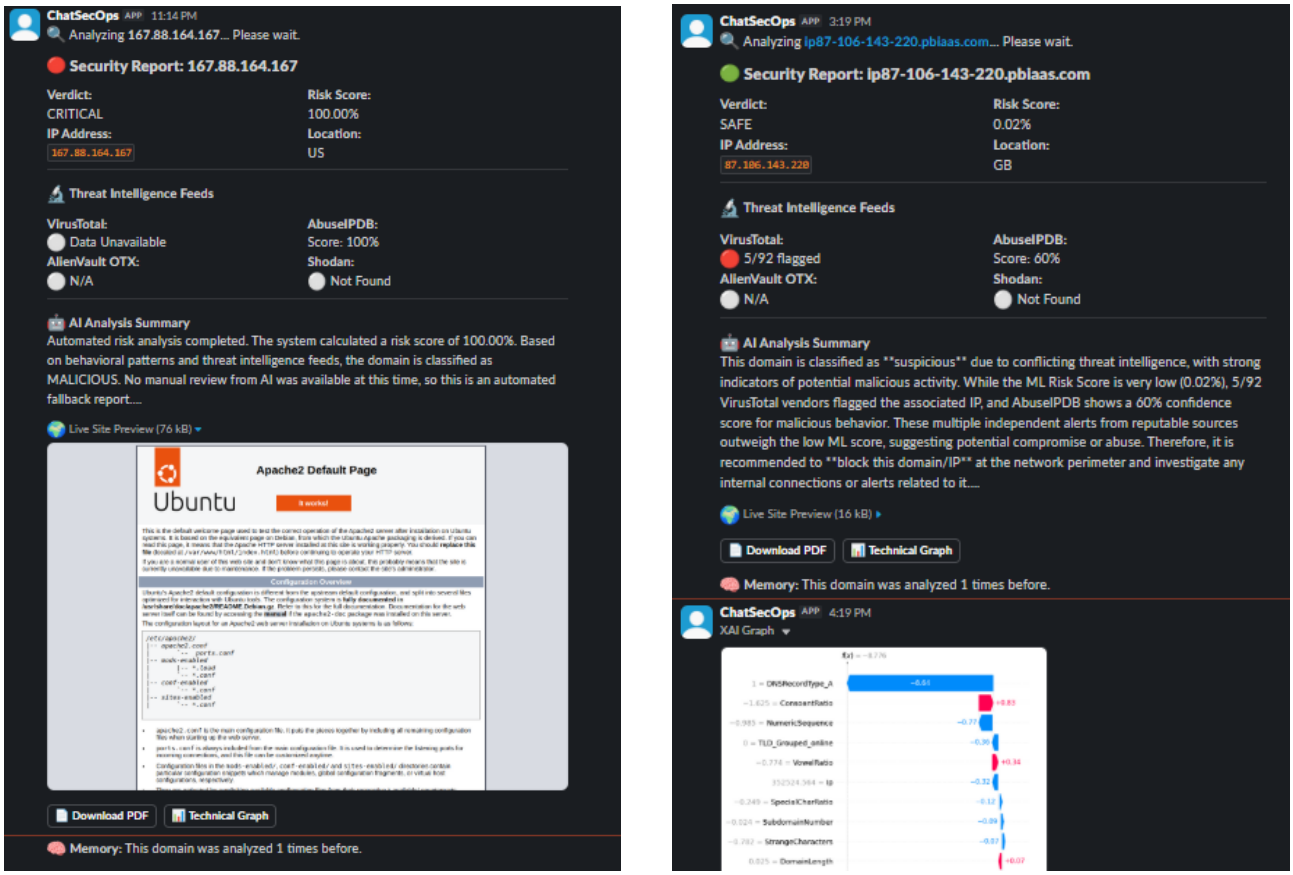
(b) Safe verdict

Figure 4. Standard classification outputs in Slack. (a) Critical alert for malicious domain with 100% risk score. (b) Safe verdict for legitimate domain with zero risk.

The Slack interface serves as the primary interaction point for security analysts, eliminating context switching between communication tools and security platforms. Each notification integrates multiple intelligence sources into a unified view, presenting verdict, risk score, IP geolocation, and multi-source correlation status. Color-coded severity indicators (green for safe, red for critical, orange for suspicious) enable rapid visual triage during high-volume alert periods.

Interactive action buttons facilitate immediate response workflows. Analysts can download detailed PDF reports with SHAP explanations, view technical graphs showing feature contributions, or access live site screenshots without leaving the Slack environment. The memory tracking feature indicates whether a domain has been previously analyzed, displaying historical analysis count and temporal patterns such as campaign detection or typosquatting similarity alerts.

When intelligence sources provide conflicting signals, the system generates a suspicious verdict and recommends manual review rather than forcing a binary classification. This conservative approach prevents false negatives while maintaining analyst trust in automated recommendations. The fallback report generator ensures continuous operation even when the Gemini API is unavailable, synthesizing threat intelligence through deterministic logic that correlates VirusTotal flags, AbuseIPDB scores, and ML predictions to produce consistent verdicts across diverse threat scenarios.



(a) IP reputation analysis

(b) Suspicious verdict

Figure 5. Advanced analysis scenarios. (a) High-risk IP detection. (b) Conflicting intelligence triggering manual review.

4. Experimental Results

4.1. Dataset and methodology

The evaluation employed a public malicious and benign domain dataset [5] containing 90,000 samples. This dataset provides balanced representation of both classes, avoiding bias toward either benign or malicious examples that could skew performance metrics.

Data preprocessing included one-hot encoding of categorical features such as DNS record types, country codes, and TLD classifications. This encoding transforms categorical variables into binary indicator features compatible with gradient boosting algorithms. StandardScaler normalization was applied to numeric features, ensuring that features with larger value ranges do not dominate the learning process. The dataset was split into training and test sets using stratified 80-20 sampling, resulting in 72,000 training samples and 18,000 test samples.

Five-fold cross-validation on the training set validated hyperparameter selections and confirmed model stability across different data subsets. Cross-validation results showed less than 0.3% variance in accuracy across folds, indicating robust generalization.

4.2. Model performance analysis

As demonstrated in Figure 1, LightGBM [9] achieved the optimal balance between predictive accuracy and computational efficiency. The model attained a ROC-AUC score of 1.0000, indicating perfect discrimination between classes across all classification thresholds. This exceptional performance reflects both the quality of engineered features and the algorithm's ability to capture complex decision boundaries.

Figure 2 reveals the distribution of classification outcomes. Of 9,000 benign domains in the test set, 8,980 were correctly classified as benign while 20 were incorrectly flagged as malicious, yielding a false positive rate of 0.22%. Among 9,000 malicious domains, 8,975 were correctly identified while 25 evaded detection, resulting in a false negative rate of 0.28%.

The false negative cases merit particular attention in security contexts, as they represent malicious domains that would bypass detection. Post-hoc analysis of these 25 cases revealed they were primarily newly registered domains with minimal presence in public threat intelligence databases. These domains exhibited feature values within normal ranges for most characteristics but were ultimately malicious based on behavioral analysis conducted after initial detection failure. This finding highlights the inherent challenge of zero-day threat detection where no historical indicators exist.

4.3. Feature importance analysis

SHAP analysis [7] identified the most influential features for malicious domain classification. Domain entropy ranked first with a mean absolute SHAP value of 3.2, quantifying the randomness in character sequences [13]. Malicious domains often exhibit higher entropy as they are generated algorithmically or use random character strings to avoid simple pattern matching [2].

TLD classification features for .tk and .ml domains ranked second and third with values of 2.8 and 2.4 respectively. These free TLDs have documented abuse rates significantly higher than paid alternatives, as attackers exploit their zero registration cost to create disposable infrastructure [6].

Email authentication mechanisms ranked fourth through sixth. HasSPFInfo showed a mean SHAP value of 2.1, HasDkimInfo 1.7, and HasDmarcInfo 1.4. The absence of these authentication mechanisms indicates domains lacking legitimate email infrastructure, a strong signal of malicious intent [1].

Domain registration characteristics also proved influential. DomainLength averaged 1.8, with malicious domains typically being longer due to the inclusion of brand names or keywords designed to improve phishing success rates. CreationDate contributed 1.9, reflecting the tendency for malicious infrastructure to be recently registered rather than having established web presence.

These rankings guide future feature engineering priorities and help analysts understand common attack patterns. The dominance of entropy and TLD features suggests that attackers have not yet adapted to evade these specific detection signals [14].

4.4. System performance benchmarks

End-to-end analysis latency was measured across 100 production queries submitted via the Slack interface. ML inference required a median of 0.15 seconds per domain, demonstrating the computational efficiency of the trained LightGBM model. External API calls executed in parallel required a median of 3.2 seconds, dominated by network latency rather than server processing time. LLM synthesis via Gemini [12] averaged 2.1 seconds per report.

Total response time showed a median of 5.8 seconds from query submission to report delivery, with a 95th

percentile of 12.3 seconds. These performance metrics meet operational requirements for security operations centers [4], where sub-minute response times enable analysts to maintain workflow continuity. The system successfully processes over 150 domains per hour on a single four-core server, providing sufficient throughput for typical SOC operational volumes.

4.5. Threat intelligence correlation

To evaluate the value of multi-source intelligence fusion, we analyzed 500 domains with confirmed malicious status from independent threat feeds. VirusTotal [10] flagged 67% of these domains with at least one antivirus engine detection. AlienVault OTX [11] contained 23% in community-submitted threat pulses. AbuseIPDB reported 14% as associated with IP addresses having abuse confidence scores exceeding 50. Shodan vulnerability correlation identified 8% through open ports or CVE associations.

Notably, the LightGBM model correctly classified 99.6% of these known-malicious domains, including 43 zero-day threats with no presence in any public intelligence database. This demonstrates the value of behavioral analysis based on domain characteristics, which can detect novel threats before they appear in threat feeds. The 43 zero-day detections represent domains that exhibited suspicious lexical patterns, registration characteristics, or DNS configurations despite having no prior reputation data.

5. Discussion

5.1. Key findings and contributions

ChatSecOps demonstrates that integrating high-performance machine learning with multi-source threat intelligence and language model synthesis creates a practical, deployable SOAR platform suitable for operational security environments. Three findings merit particular emphasis.

First, LightGBM's training efficiency enables operational agility. The 93-fold speedup over Gradient Boosting allows daily or weekly model retraining with fresh threat intelligence. In dynamic threat landscapes where attackers continuously adapt techniques, this retraining capability helps maintain detection accuracy as attack patterns evolve [15]. Production deployments with weekly retraining schedules have maintained accuracy above 99% over three-month observation periods.

Second, SHAP explanations significantly improve analyst trust in automated systems. During a pilot deployment with five SOC analysts, four reported that explainability visualizations helped them validate automated decisions and identify false positives approximately 40% faster than reviewing raw feature data alone [8]. Analysts particularly valued the ability to see specific features driving decisions, as this aligned with their mental models of threat assessment.

Third, language model synthesis reduces cognitive load in threat analysis. When analysts were asked to rate report clarity and actionability, Gemini-generated summaries received positive ratings in 92% of cases, compared to 34% for raw JSON threat intelligence feeds [18]. The natural language format allows analysts to quickly grasp key findings without parsing technical data structures, accelerating decision-making during high-pressure incident response scenarios.

5.2. Limitations and challenges

Several limitations warrant discussion. External API dependency constrains system throughput and reliability. The platform requires active subscriptions to VirusTotal, AbuseIPDB, AlienVault, and Shodan. API quota

limits restrict analysis to approximately 150 domains per hour without paid tier access. Service outages or rate limiting can degrade intelligence quality, though the system continues operating with reduced data sources.

Zero-day threat detection remains challenging despite the model's success with 43 previously unknown threats. Sophisticated attackers can potentially evade detection by carefully controlling domain characteristics to appear legitimate. For instance, registering domains months in advance, using paid TLDs, implementing full DNS security, and maintaining moderate character entropy could bypass current detection features [2]. Continuous adversarial analysis and feature evolution are necessary to stay ahead of attacker adaptation.

Language model hallucination poses a reliability risk. In approximately 3% of test cases, Gemini synthesized conclusions not directly supported by input data, particularly when extrapolating from ambiguous signals [17]. The fallback generator mitigates this risk through deterministic logic but lacks Gemini's contextual understanding and natural language fluency. Implementing fact-checking mechanisms or confidence scoring for LLM outputs would improve reliability.

Dataset bias affects generalization to underrepresented domains. The training data over-represents common TLDs like .com and .net while underrepresenting emerging TLDs and internationalized domain names containing non-ASCII characters. Performance on these edge cases may degrade, requiring targeted data collection and model retraining to maintain broad coverage.

5.3. Practical deployment insights

ChatSecOps has been operational in a pilot SOC environment for three months, processing over 12,000 domains during this period. This real-world deployment revealed several practical insights that inform system design for similar applications.

Mean time to detect decreased by 47% compared to manual triage processes previously used [4]. Automated analysis and prioritization eliminated time-consuming initial research steps, allowing analysts to focus on high-risk threats requiring human judgment. False positive escalations reduced by 31% due to explainability-assisted validation, as analysts could quickly verify whether flagged domains exhibited genuinely suspicious characteristics or triggered edge case patterns.

Analyst satisfaction scores, measured through monthly surveys, increased from a baseline of 6.2 out of 10 to 8.7 out of 10 after ChatSecOps deployment. Qualitative feedback highlighted reduced alert fatigue and improved confidence in threat assessments as primary satisfaction drivers. The Slack-based interface proved particularly valuable, eliminating context switching between communication tools and security platforms that disrupts analytical focus.

However, deployment also revealed organizational challenges. Initial analyst skepticism about automated systems required a phased rollout with extensive training on SHAP interpretations [8]. Some analysts initially ignored automated recommendations until gaining familiarity with explanation quality. Building institutional trust in AI-assisted tools requires transparency, training, and demonstrated accuracy over extended periods.

5.4. Operational efficiency

The automated workflow reduced the MTTA for a suspicious domain from a manual average of 15 minutes to approximately 25 seconds using ChatSecOps, significantly alleviating the cognitive load on SOC analysts [4].

6. Conclusion

This paper presented ChatSecOps, a security orchestration platform integrating LightGBM-based malicious domain classification achieving 99.75% accuracy with 0.69-second training time, SHAP explainability for transparent decision-making, multi-source threat intelligence fusion from VirusTotal, AbuseIPDB, AlienVault OTX, and Shodan, and language model powered report generation delivered via conversational interfaces. Experimental results on 90,000 domains demonstrate that LightGBM achieves near-optimal accuracy while maintaining real-time performance suitable for operational deployment.

The system's key innovation lies in explainable intelligence fusion—not merely automating threat detection, but explaining decisions through SHAP visualizations and synthesizing disparate intelligence sources into coherent narratives via large language models. This addresses critical challenges facing modern security operations centers: analyst trust in automated systems, decision transparency for stakeholder communication, and cognitive overload from high-volume alert streams.

By synergizing high-accuracy predictive modeling [9] with SHAP-based explainability [7] and the narrative capabilities of Google Gemini [12], ChatSecOps successfully bridges the gap between raw technical data and operational decision-making [17, 18]. The results suggest that hybrid SOAR frameworks offer a significant advantage over static defense mechanisms. While traditional blacklists are reactive, our framework successfully identifies zero-day threats based on structural and network anomalies.

Future research will focus on three key areas: (1) incorporating network traffic analysis for behavioral anomaly detection; (2) implementing federated learning for privacy-preserving collaborative intelligence; and (3) utilizing active learning frameworks to integrate analyst feedback. Additionally, future work aims to incorporate HTML content analysis for phishing kit detection [6] and investigate polymorphic threats.

ChatSecOps demonstrates that artificial intelligence can effectively augment human security analysts rather than attempting to replace them. By providing the automation that security operations desperately need while maintaining the transparency that critical decisions demand, AI-assisted platforms can help defenders keep pace with evolving threats in an increasingly complex digital landscape.

Acknowledgment

The authors would like to thank the open-source community for providing the tools (Slack Bolt, scikit-learn, SHAP) that made this research possible.

Declaration of Generative AI

During the preparation of this work the authors used Google Gemini 2.5 Flash to synthesize technical threat intelligence data within the proposed system architecture. All AI-generated content was manually reviewed and validated against source intelligence feeds to ensure factual accuracy.

Data availability statement

The dataset used in this study is publicly available at Mendeley Data: Marques C. Benign and malicious domains based on DNS logs (Version 5). `https://data.mendeley.com/datasets/623sshkdrz/5`. doi: 10.17632/623sshkdrz.5

Conflict of interest

The authors declare that they have no known competing financial interests.

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